

What are logic gates?

Electronic circuits that can be used to implement the most elementary logic expressions (Boolean expressions)

Basic logic gates - AND, OR, NOR, NAND, NOT

Truth table - lists all possible combinations of input binary variables

$$\text{AND} \rightarrow Y = A \cdot B \quad \text{NOR} \rightarrow Y = (\overline{A+B})$$

$$\text{OR} \rightarrow Y = A+B \quad \text{NAND} \rightarrow Y = (\overline{A \cdot B})$$

$$\text{NOT} \rightarrow Y = \overline{A}$$

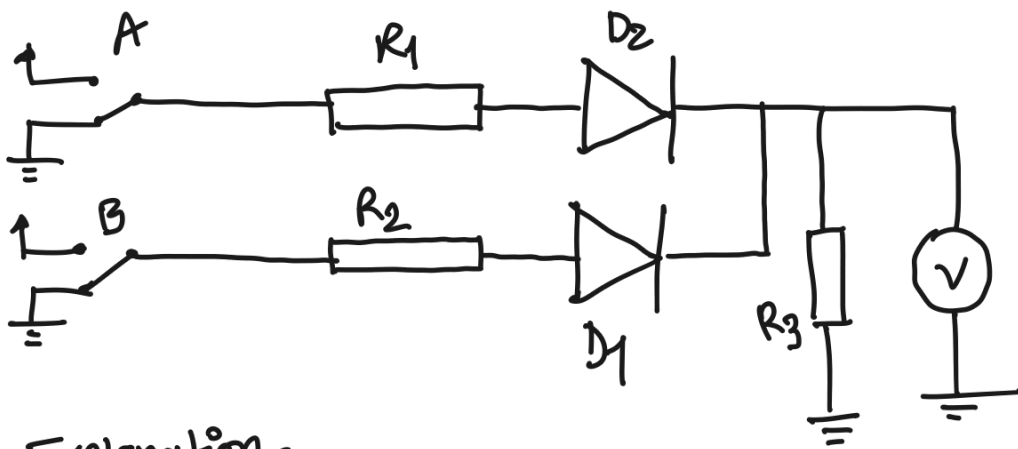
Logic families - There are a variety of circuit configurations or more appropriately - various approaches used to produce different types of digital integrated circuit, each fundamental approach is called a logic family

Types -

- Bipolar Devices
- Diode logic (DL)
 - Resistor transistor logic (RTL)
 - Diode transistor logic (DTL)
 - Transistor transistor logic (TTL)
 - Emitter coupled logic (ECL) / Current mode logic (CML)
 - Integrated injection logic (I²L)
- Also - MOS family, PMOS, NMOS, CMOS family, $\rightarrow$ MOS devices exist
Bi-MOS family exist

✓ DL OR gate

(Not shown in class but for understanding)

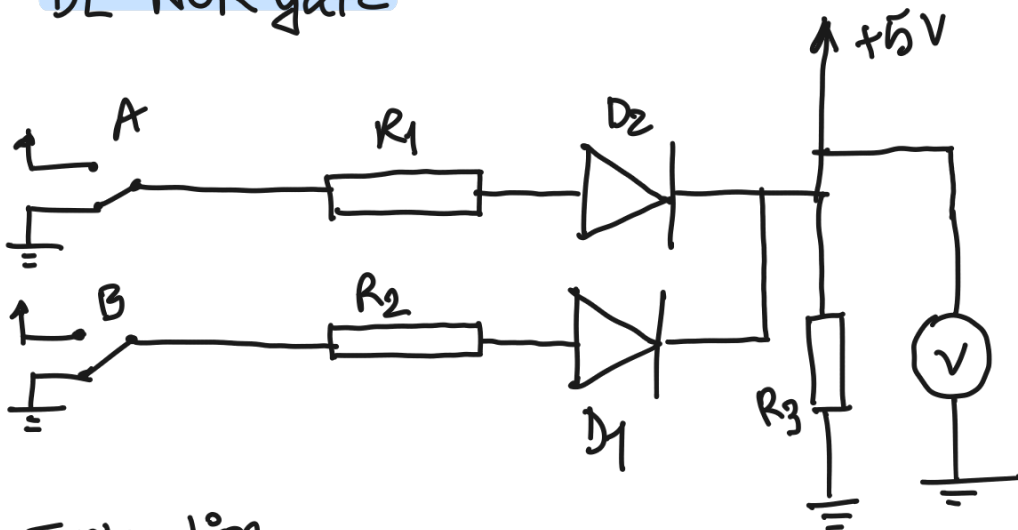


<u>A</u>	<u>B</u>	<u>Y</u>
0	0	0
0	1	1
1	0	1
1	1	1

Explanation -

If either A or B is on, output is 1.

✓ DL NOR gate



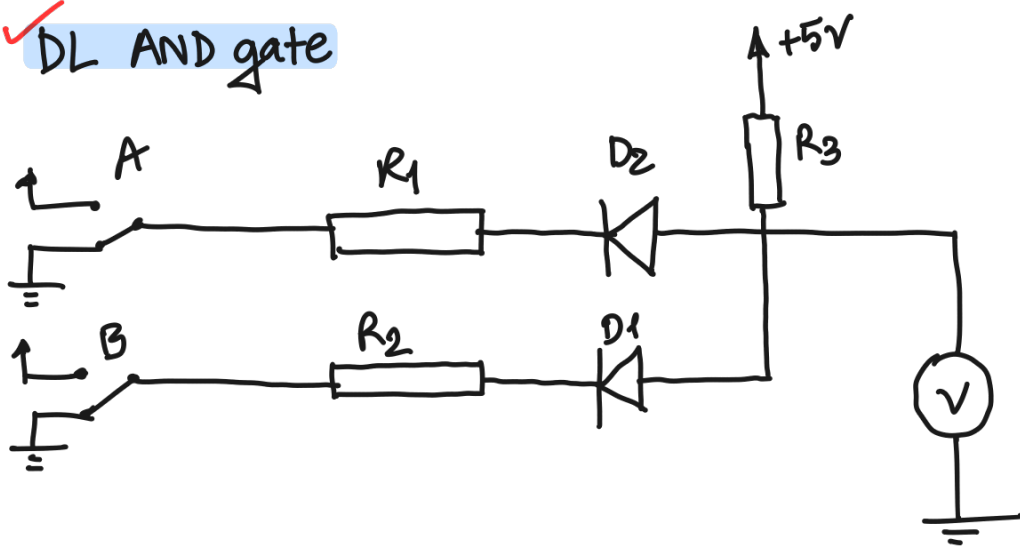
<u>A</u>	<u>B</u>	<u>Y</u>
0	0	1
0	1	0
1	0	0
1	1	0

Explanation -

$A=0$ $B=0$, no potential difference between A, B, ground current won't flow through those paths, so output 1.

$A=0$, $B=1$, potential difference makes current flow to ground, not the output

✓ DL AND gate



<u>A</u>	<u>B</u>	<u>Y</u>
0	0	0
0	1	0
1	0	0
1	1	1

Explanation -

When $A=0, B=1$
(0v) (5v)

For current to flow through $D_1 \rightarrow 5.7V$ needed (0.7V higher)
Since 5V supplied, D_1 off

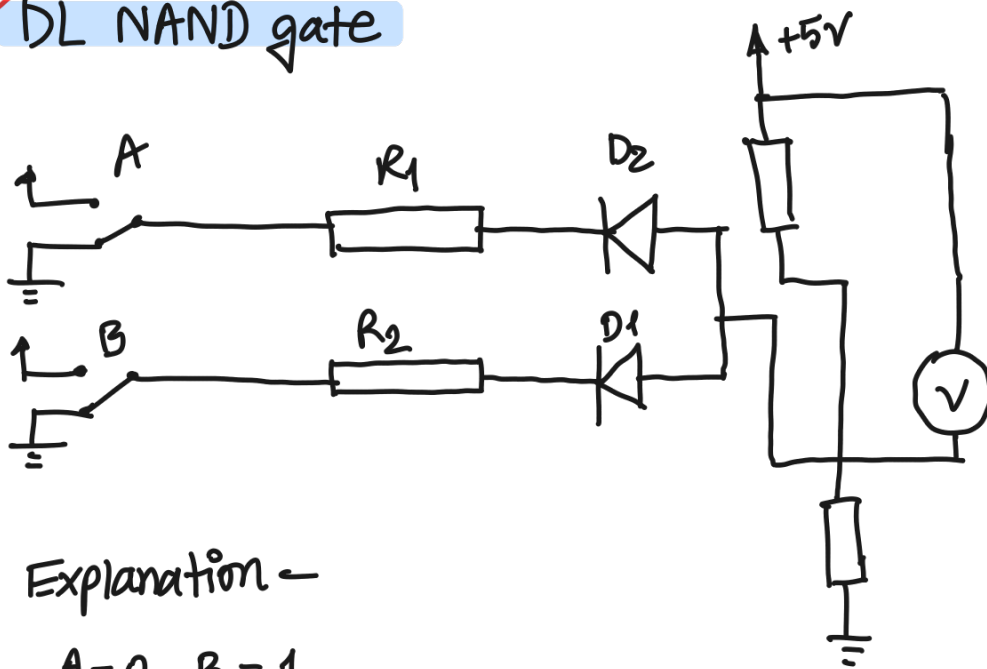
For D_2 , 0.7V needed, since 5V supplied, D_2 on
output $\rightarrow 0V$ since $A \rightarrow 0V$

When $A=1, B=1$
(5v) (5v)

For current to flow through $D_1 \rightarrow 5.7V$ needed (0.7V higher)
Since 5V supplied, D_1 off

For D_2 , 5.7V needed, since 5V supplied, D_2 off
output $\rightarrow 5V$ since current can only pass through the other path

✓ DL NAND gate



<u>A</u>	<u>B</u>	<u>Y</u>
0	0	1
0	1	1
1	0	1
1	1	0

Explanation -

$A=0, B=1$

D_1 to be on $\rightarrow 5.7V \rightarrow$ stays off

D_2 " " " $\rightarrow 0.7V \rightarrow$ stays on ($A=0V$)
0V potential

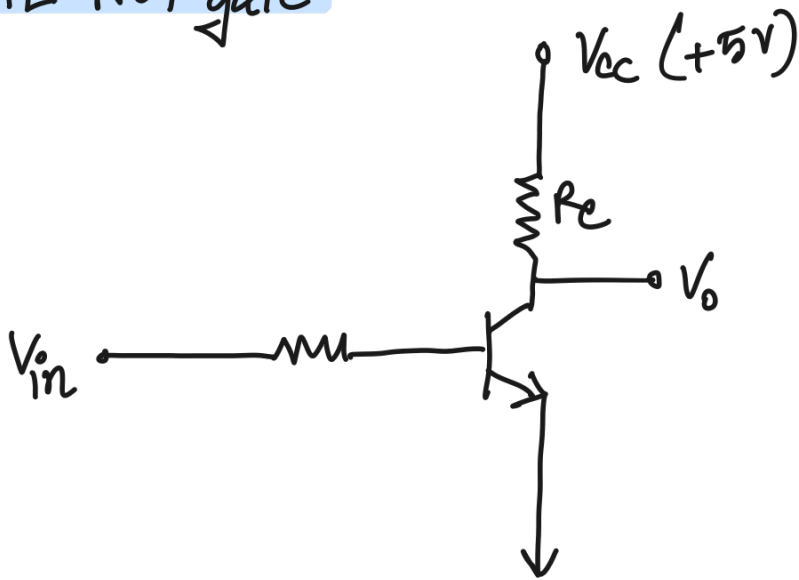
Through output current passes since potential difference exists (5V, 0V).

$A=1, B=1$

D_2, D_1 both off $\rightarrow$ either through resistances or the output

passes through the resistances, output
0

✓ RTL NOT gate



V_{in}	V_o
0	1
1	0

Explanation -

$V_{in} = 0$, Base $\rightarrow 0V$ cut off mode

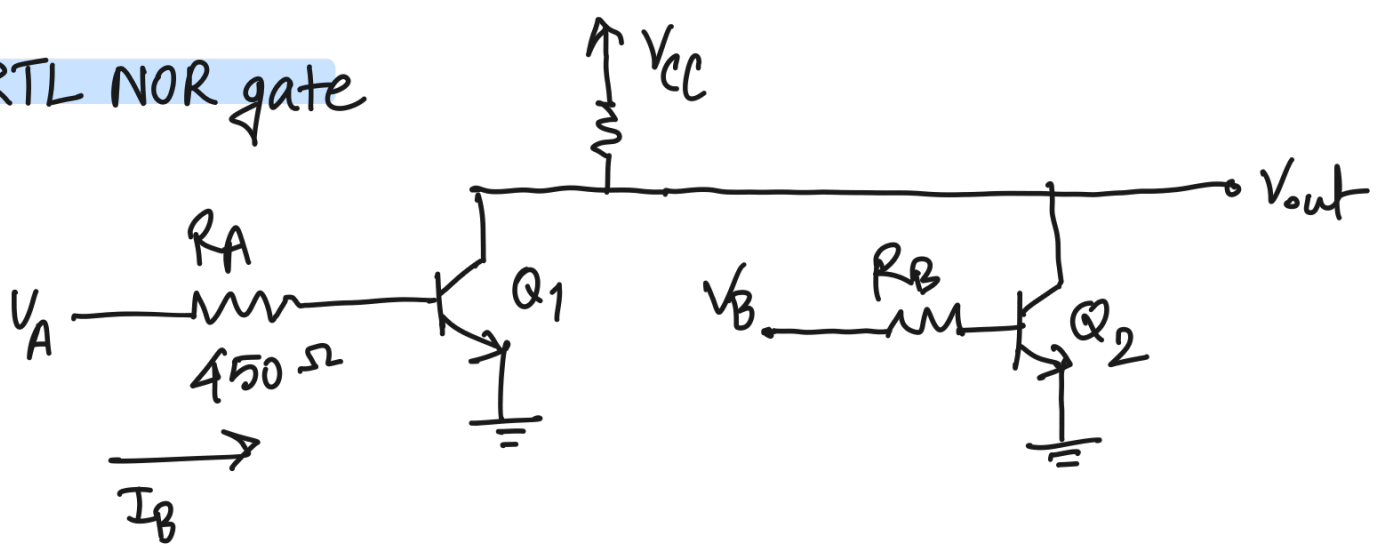
$V_o = V_{cc} = 5V$ (1 logic) emitter-collector open circuited

$V_{in} = 1$ (5V), Base $\rightarrow$ current supplied

$V_o = 0.2V$ (0 logic)
(V_{CE})

Emitter to collector potential drop happens
 V_{CE}

RTL NOR gate



Explanation -

$V_A = V_B = 0$, Q_1, Q_2 off

Current from V_{CC} flows to V_{out}
logic - 1

One input $\rightarrow 1$

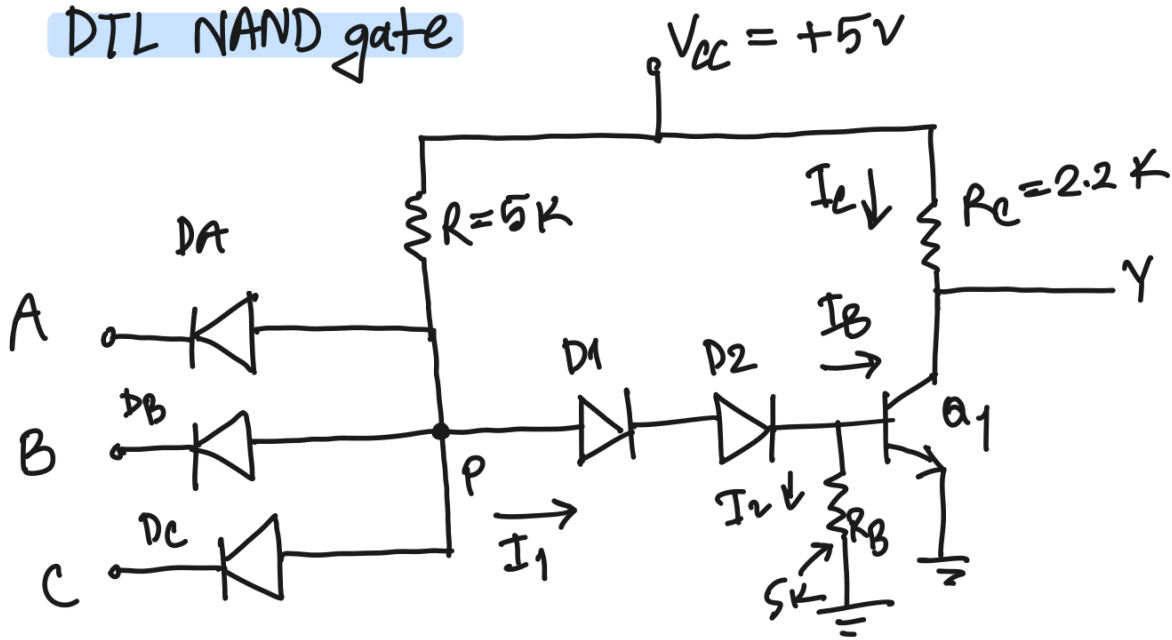
One of the transistors in saturation mode

$V_{CE} = 0.2\text{ V}$, $V_{out} = 0.2\text{ V}$ (0 logic)

Both 1 $\rightarrow$ both node have $V_{CE} = 0.2\text{ V}$
 $V_{out} = 0.2\text{ V}$ (0 logic)

$\underline{V_A}$	$\underline{V_B}$	$\underline{V_{out}}$
0	0	1
1	0	0
0	1	0
1	1	0

DTL NAND gate



Assumptions -

$$V_{BE} = 0.8V \text{ (sat.)}$$

$$V_{BE} = 0.5V \text{ (cut.)}$$

$$V_{CE} = 0.2V \text{ (sat.)}$$

$$\text{Diode voltage drop} = 0.7V$$

A, B, C or A, B both can be explained using same logic -

Explanation -

$$A=0, B=0$$

DA, DB Forward bias

D1, D2 Reverse bias $\rightarrow$ Q1 off

$$Y \rightarrow V_{CC}$$

$$A=0, B=1$$

DA forward bias, DB reverse

D1, D2 reverse $\rightarrow$ Q1 off

$$Y \rightarrow V_{CC}$$

$$A=1, B=1$$

DA, DB reverse bias

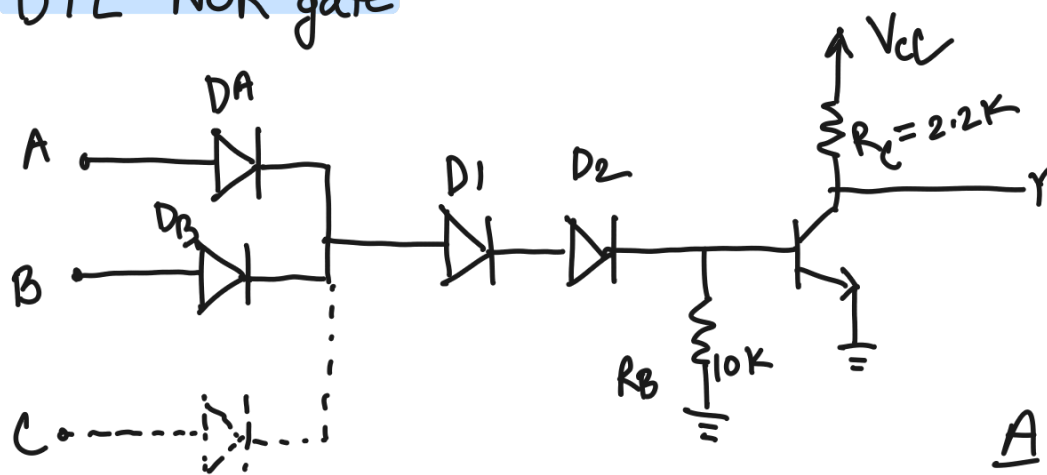
D1, D2 forward bias

Q1 on, $Y = V_{CE} = 0.2V$

(logic 0)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

DTL NOR gate



Explanation -

$A=0, B=0$

D_A, D_B reverse bias

D_1, D_2 "

$Y = V_{CC}$ (logic 1)

<u>A</u>	<u>B</u>	<u>Y</u>
0	0	1
0	1	0
1	0	0
1	1	0

$$0.7 + 0.7 + 0.7 + 0.8 = 2.9 \text{ V}$$

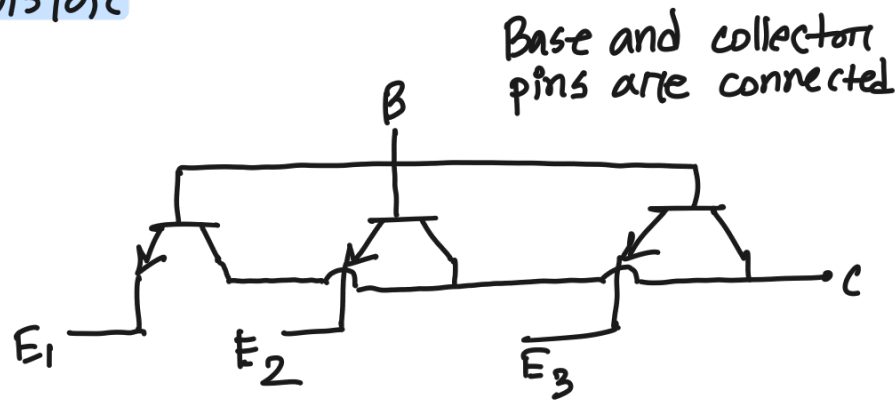
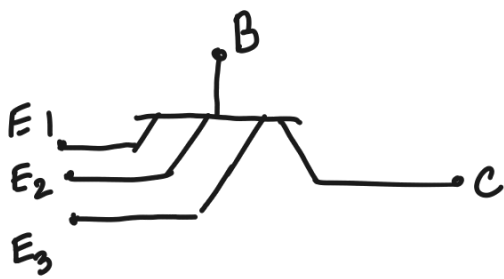
$A=1, B=0$

D_A forward bias, D_B reverse bias

D_1, D_2 forward bias since current flows

$Y = V_{CE} = 0.2 \text{ V}$ (logic 0)

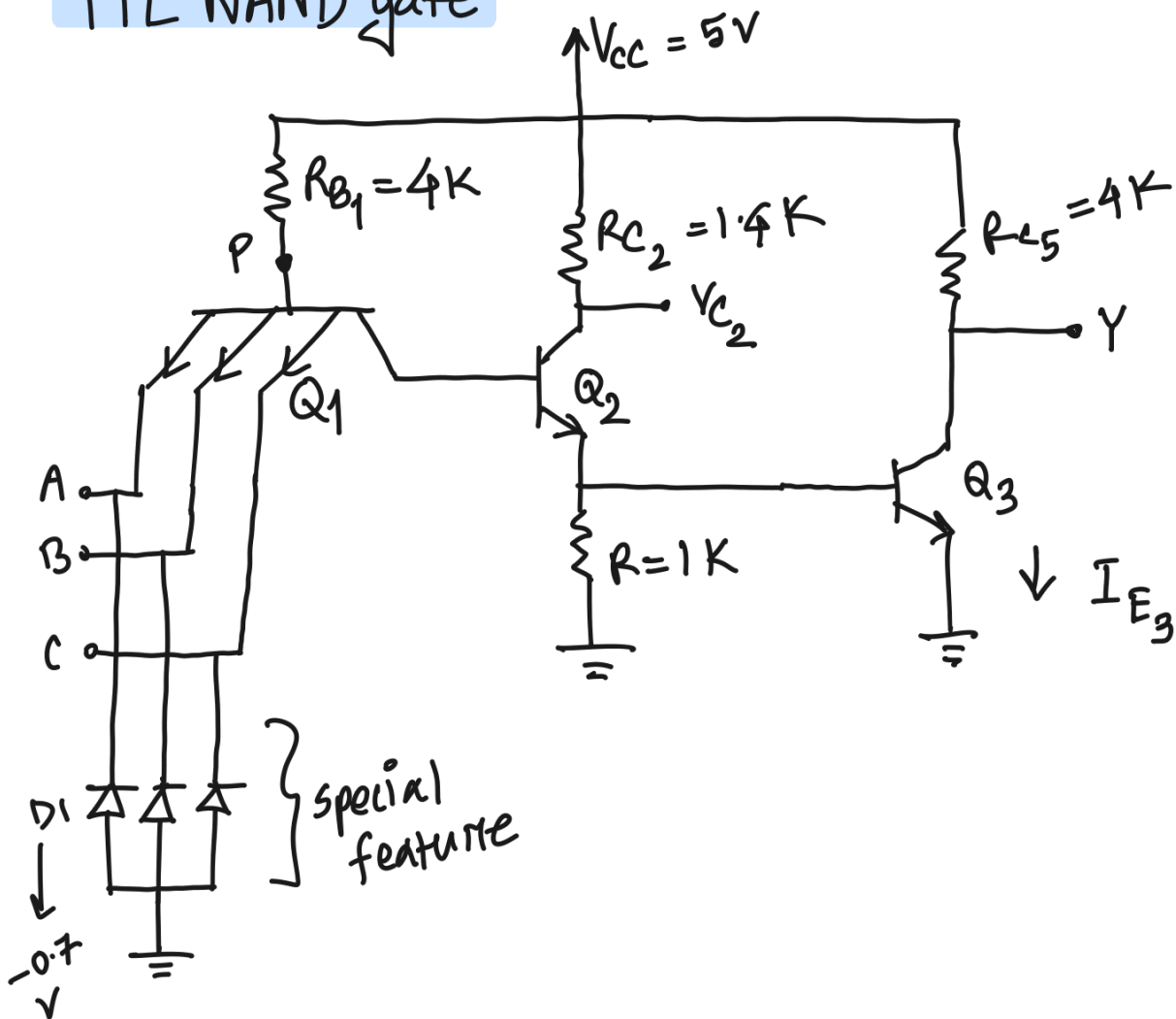
TTL multi emitter transistor



$$E_1 = E_2 = 5V, E_3 = 0V; B = 0.7V$$

At least one of the emitters $\rightarrow$ Low, one of the transistors are forward biased (entire thing turned on)

TTL NAND gate



<u>A</u>	<u>B</u>	<u>C</u>	<u>Y</u>
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Case-1

$V_A \rightarrow \text{low}$

$V_B \rightarrow \text{high}$

$V_C \rightarrow \text{high}$

for logic low, gate transistor is saturated and output voltage is 0.2 V

for logic high, gate transistor turned off and output 5 V (V_{CC})

A in low state - diode D_A conducts

$$V_p = 0.2 + 0.7 = 0.9 \text{ V}$$

at least one of the inputs Low $\rightarrow D_1$ ON but not D_2
output logic high

$$D_1 = D_2 = 0.7, V_{BE} = 0.8 \text{ V (sat.)}$$

$A = 0.2 \text{ V}, B = C = 5 \text{ V} \rightarrow \text{junction turned on}$

$$V_p = 0.2 + 0.7 = 0.9 \text{ V}$$

other transistors will not turn on as collector = 0.7 V

(not more than that)
 $\rightarrow$ needs 2.1 V to turn on
($0.7 + 0.7 + 0.7$)

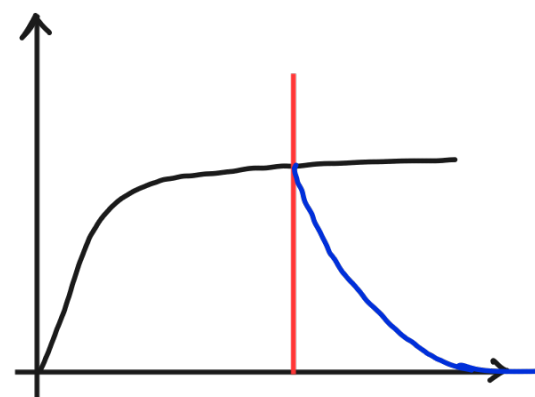
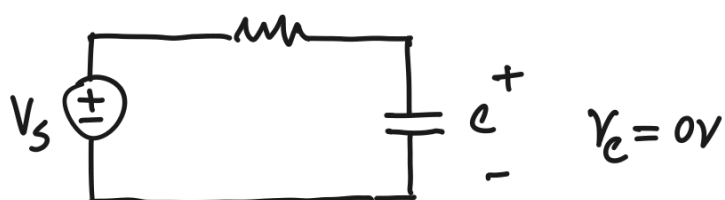
all inputs are logic high $\rightarrow$ reverse biased

$$\text{voltage needed} = 0.7 + 0.8 + 0.8 = 2.3 \text{ V}$$

When all inputs High, Q_3, Q_2 saturation mode

Time constant τ

$$\tau = RC$$



with diodes, $D_1 \Rightarrow V_A - 0.7 \text{ V}$

Math In the circuits of TTL NAND gate, if all the inputs are in high condition, then determine the emitter current of Q_3 .

solution: $V_A = V_B = V_C = 5 \text{ V}$

Q_1 = reverse active mode
BE junction reverse biased
BC " forward "

Q_2, Q_3
saturation

$$V_P = V_{BC_1} + V_{BE_2} + V_{BE_3} = 0.7 + 0.8 + 0.8 = 2.3 \text{ V}$$

$$I_{B_1} = \frac{V_{CC} - V_P}{R_{B_1}} = \frac{5 - 2.3}{4K} = 0.675 \text{ mA}$$

(through R_{B_1})

$$I_{E_1} \approx 0V \quad I_{E_1} = I_{C_1} + I_{B_1}$$

$$0 = I_{C_1} + I_{B_1}$$

$$I_{C_1} = -0.675 \text{ mA} \rightarrow I_{B_2} \text{ (current through base of } Q_2)$$

$$V_{BE_3} = V_{B_3} = 0.8V$$

$$\frac{I_R}{I_{1K}} = \frac{V_{B_3}}{1K} = \frac{0.8}{1K} = 0.8 \text{ mA}$$

$$V_{C_2} = V_{B_3} + V_{CE_2} = 0.8 + 0.2 = 1V$$

$$I_{C_2} = \frac{V_{CC} - V_{C_2}}{1.4K} = \frac{5 - 1}{1.4K} = 2.86 \text{ mA}$$

$$I_{E_2} = I_{B_2} + I_{C_2} = (0.675 + 2.86) \text{ mA} = 3.53 \text{ mA}$$

$$I_{B_3} = I_{E_2} - I_{1K} = 3.53 - 0.8 = 2.73 \text{ mA}$$

$$I_{C_3} = \frac{V_{CC} - 0.2}{4K} = \frac{5 - 0.2}{4K} \overset{\text{sat.}}{=} 1.2 \text{ mA}$$

$$I_{E_3} = I_{B_3} + I_{C_3} = 3.43 \text{ mA}$$

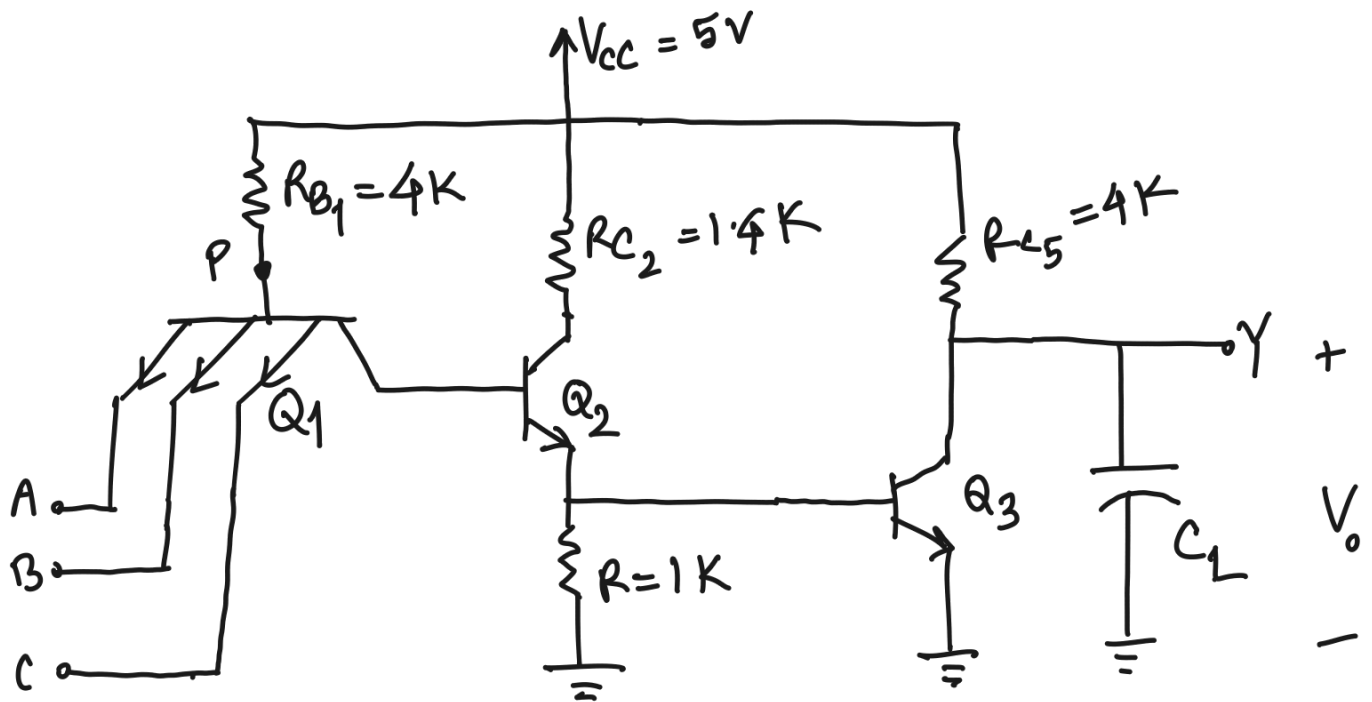
TTL : Improvement in output stages

Load capacitance is the total capacitance that the output of a logic gate must charge or discharge during a logic transition (high $\leftrightarrow$ low)

This capacitance introduces delays in signal transitions, limiting switching speed.

Sources of load capacitance :

- Every TTL input pin behaves like a small capacitor due to the base emitter junction of the input BJT (in reverse biased condition $\rightarrow$ capacitance due to depletion region acting as di-electric between two charge regions)
- The physical wires or PCB traces connecting gates act like capacitors between trace to GND, trace/signal line to adjacent signal lines



❑ TTL output must charge and discharge load capacitance every time logic level changes.

❑ If passive pull up resistor R_C is used for this purpose $\rightarrow$ time constant $R_C C_L$ is high, causing long delay time when logic level experience a transition

Totem pole configuration

